

Programming Domains

Computers have been applied to different areas from controlling nuclear power plants to providing video games in mobile phones. Because of this great diversity in computer use, PLs with very different goals have been developed. Some of the few areas of computer applications & the associated languages include:

a) Scientific Applications

In the early 1940s and 1950s, computers were invented for scientific applications.

Typically, the scientific applications of that time use relatively simple data structures but required large number of floating point arithmetic computations.

At that time the most common data structure that were used:

- Arrays & Matrices
- Control Structures - Loop & selections

The early high level language invented for scientific applications were designed to provide for those needs.

FORTRAN - FORmula TRANslation

- ALGOL GO first version to address scientific concerns

b) Business Applications.

The use of computers for business applications began in the 1950s.

Special computers were developed for this purpose along with special languages.

- COBOL Common Business Oriented Language

still being used
because of its
efficiency

- COBOL is still the most commonly used language for this applications.
- Business languages are characterised by facilities for producing elaborate reports.
- Precise ways of describing & string decimal numbers.
- Character data
- Ability to specify decimal arithmetic operations.

c) Artificial Intelligence

- AI is a broad area of computer applications; characterised by the use of symbolic rather than numeric computations.
- This means that symbols consisting of names rather than numbers are manipulated.
- This kind of programming is more conveniently done with linked list rather than Arrays.
- The first widely used programming lang for this application.
- LISP (1959-1990)
- Logic Prog - Prolog (1970s)

d) Systems Programming

The OS & the programming support tools of a computer system are collectively known as its system's software.

System SW is used almost continuously & so it must be efficient.

It must also have low level features that allow the software interfaces to external devices to be written.

Prog lang that fall into this domain:

- IBM - PL/I ^{interface} changed later to PL/S - system all originating from BLISS (above assembly lang)
- Burroughs - Extended ALGOL
- currently - C/C++

c) Web Software

- The World Wide Web (WWW) is supported by an eclectic collection of languages ranging from markup languages such as HTML, CSS, JAVASCRIPT
- Because of the pervasive need for dynamic web content some compilation capability is often included in the tech of content presentation

Domains $\rightarrow$ This functionality can be embedded in the such code is often in the form of a scripting language such as JavaScript & PHP

Language Evaluation Criteria

- a) Readability how easy it is for a given programming lang can be read and understood

One of the most important criteria of judging a prog^l is the ease with which programs can be read & understood

Key indication readability:

- overall simplicity: of a programming lang strongly affects its readability.

A language with a large^{no} of basic constructs ^{is} more difficult to learn than one with a smaller number.

- b) - Orthogonality



combine small construct
e.g. if-else

It is a ~~read~~ relatively small set of primitive construct can be combined in a relatively small^{no} ways to build the control in data structures of the lang
E.g.

Data Types
The presence of adequate facilities for defining data types & data structures in a language is another of significance as to readability.

- Syntax Design (rules given to programming lang)

- The syntax or form of the elements of a language has significant effect on the readability of programs.
- Syntax is defined by ^① special words $\rightarrow$ (the keywords that are used in a programming lang that is understood by the compiler)
- ② Form & meaning $\rightarrow$ this

b) Writability

It is a measure of how easily a language can be used to create programs for a chosen problem domain.
Most of the language characteristics that affect readability also affects writability.

- i) Simplicity & orthogonality (same definition) *
- ii) Support for abstraction

Abstraction means the ability to define and then use complicated structures or operations in ways that allow many of the details to be ignored.

- iii) Expressivity - easier it is to express, the easier the writability.
- Expressivity in a language can refer to several different characteristics, such as APL it means language has there relatively are very powerful operators that allow a great deal of computation accomplish with a very small program.
- Expressivity commonly means a language has relatively convenient rather than cumbersome ways of specifying computations
e.g. `counter = counter + 1 ;` or `counter = counter + 1 ;`

d) Reliability

A program is said to be reliable if it performs to its specifications under all conditions

- i) Type checking is simply checking/testing for type errors in a given program, either by the compiler or during program execution.

Run-time type checking is more expensive
Compile-time type is better. e.g. Java

- ii) Exception Handling the ability of a program to intercept a run-time errors as well as other unusual condition detected by the program, take corrective measures and then continue is an obvious aid to reliability. This is called exception handling such as Java, C#, C++, ARDER

- iii) Aliasing. This is an instance where memory location has more than one name.

It is having two or more distinct names that can be used to access the same memory cell.

It is now widely accepted that aliasing is a dangerous feature in a programming language.

Most programming languages allow some kind of aliasing e.g. use of pointers (C, C++ ..)

* Readability & writability indicators affect also reliability.

Influences on language Design

Date
M T W T F S S

a) Computer Architect Architecture

The basic architecture of computers has had a profound effect on language design.

Most of the popular languages of the past 50 years has been designed around the prevalent computer architecture called **Von Neumann Architecture**.

These languages are called imperative languages.

b) Programming Design Methodologies

In the late 1960's and early 1970's an intense analysis began in large part by the structured programming movement of both the software development process & programming language design.

data
oriented
↓
Java (object
oriented)

Language categories (assessment) with examples and what do they have in common.

Language Design Trade-offs

Level of compromise

depends on

what you are facing
(find a balance)

1. Conflict between reliability & cost e.g. exception handling will have you writing extra code

2. Conflict between writability and reliability.

3. Expressivity & Readability - APL
Writability

Implementation Methods assessment

- How compilation is implemented generally

Pure interpretation & Implementation

Hybrid Implementation

What are the good things in compilation than implementation

Programming environments like Unix, Golang, netbeans, Microsoft windows

When are these programming languages environments preferred over the other.